

HAF-EPA: Intelligent Employee Recommendation System

Project Description and Methodology

1. Project Goal

The primary objective of the HAF-EPA (Human Assignment Framework – Employee Project Allocation) project is to develop an intelligent employee recommendation system that assists organizations in identifying the most suitable employees for specific projects.

In many organizations, assigning employees to projects is often performed manually by project managers. This process can be time-consuming, subjective, and may not always result in the optimal allocation of resources. The HAF-EPA system addresses this challenge by utilizing machine learning techniques to analyze employee profiles, project requirements, historical performance, and availability information to generate data-driven recommendations.

The system aims to:

- Improve project staffing decisions.
- Reduce manual effort in employee selection.
- Increase project success rates by assigning employees whose skills align with project requirements.
- Consider multiple factors simultaneously, including technical skills, experience, availability, and historical project performance.
- Provide a scalable and automated solution for workforce allocation.

The final outcome of the project is a web-based application where users can upload a project description and receive a ranked list of recommended employees based on predicted suitability scores.

2. Data Collection and Dataset Preparation

A machine learning model is only as effective as the quality of the data used to train it. Therefore, significant effort was invested in collecting and preparing comprehensive employee and project-related datasets.

The project utilizes multiple interconnected datasets representing different aspects of organizational knowledge.

Employee Information Dataset

This dataset contains basic employee information, including:

- Employee ID
- Employee Name
- Department
- Job Position
- Years of Experience
- Employment Status

This serves as the foundational dataset for employee profiling.

Employee Skills Dataset , This dataset records the skills possessed by each employee.

Information includes:

Employee ID, Skill Name ,Skill Category , Skill Proficiency Level and Years of Experience in Each Skill

Examples of skills include:

Such as , Python, Java ,JavaScript , React, Machine Learning , Project Management

Projects Dataset, This dataset contains information about previously completed projects.

Typical attributes include:

- Project Name
- Project Category
- Project Complexity
- Project Priority
- Technology Stack

Project Skills Dataset, This dataset defines the skills required for each project.

Information includes:

Required Skills , Optional Skills , Skill Importance and Minimum Proficiency Requirements

This dataset is critical because it establishes the relationship between employee capabilities and project demands.

Employee Project History Dataset ,

Historical project participation data is used to understand employee performance trends.

Information includes:

Projects Completed , Performance Ratings, Project Outcomes , Contribution Scores

This dataset allows the model to learn from previous assignment successes and failures.

Employee Availability Dataset

Availability information is essential when assigning employees to new projects.

Attributes include:

Allocation Percentage , Available Capacity , Current Workload

Employees who are already heavily allocated receive lower availability scores.

Employee Relationship Dataset , This dataset captures collaboration information between employees.

It helps estimate:

Team Compatibility , Collaboration Effectiveness and Communication Efficiency

Such information contributes to better team formation recommendations.

Skill Reference Dataset, A master skill repository was also maintained to standardize skill names and categories across all datasets.

3. Data Cleaning and Normalization

Raw datasets often contain inconsistencies, missing values, duplicate entries, and formatting issues. Therefore, a comprehensive data normalization process was implemented before training the machine learning model.

Data Cleaning,

The following preprocessing steps were performed:

Standardization of Column Names, Column names were converted into a consistent format by:

- Removing unnecessary spaces
- Converting all names to lowercase
- Ensuring naming consistency across datasets

Missing Value Handling, Missing values were identified and handled appropriately.

Strategies included:

- Replacing empty values with defaults
- Ignoring incomplete records when necessary
- Validating mandatory fields

Duplicate Removal, Duplicate records were detected and removed to prevent bias during model training.

Data Type Conversion

Several fields required conversion into appropriate formats.

Examples include:

- Numerical fields converted to integer or float values
- Date fields converted into datetime formats
- Percentage fields standardized into numerical representations

Feature Normalization,

To ensure consistency across different data sources, normalization techniques were applied.

Examples include:

Skill Normalization, Skill names were standardized.

For example:

- "JS" and "JavaScript" were mapped consistently.
- Alternative skill names were unified.

4. Feature Engineering and Model Development

Feature engineering is one of the most important stages of any machine learning project.

The goal was to transform raw employee and project data into meaningful numerical features that the model could understand.

Employee–Project Pair Generation

Instead of training directly on employees or projects independently, the system creates employee-project pairs.

Each pair represents the possibility of assigning a particular employee to a particular project.

Feature Extraction

Several predictive features were generated.

Skill Match Features

These measure how closely an employee's skills align with project requirements.

Examples:

- Number of matched required skills
- Number of matched optional skills
- Skill coverage percentage
- Weighted skill match score

Experience Features

These capture the employee's expertise level.

Examples:

- Average experience in required skills
- Total relevant experience
- Experience-to-project complexity ratio

Performance Features

Historical performance indicators were incorporated.

Examples:

- Average project performance score
- Historical success rate
- Previous project evaluation scores

Machine Learning Model Selection

After evaluating the problem requirements, a Random Forest-based classification approach was selected.

Random Forest was chosen because:

- It handles complex feature interactions effectively.
- It performs well on structured tabular data.
- It reduces overfitting through ensemble learning.
- It provides strong predictive performance with relatively low tuning requirements.

Model Training Process

The training workflow consists of:

1. Loading normalized datasets.
2. Generating employee-project pairs.
3. Creating engineered features.
4. Constructing training labels.
5. Preparing feature matrices.
6. Training the Random Forest model.
7. Saving the trained model for deployment.

The final trained model learns patterns from historical employee-project assignments and predicts future assignment suitability.

5. Model Validation and Testing

To ensure that the model generalizes well to unseen data, a rigorous evaluation process was implemented.

Dataset Splitting

The dataset was divided into:

- Training Set (80%)
- Testing Set (20%)

The training set was used to teach the model, while the testing set was reserved exclusively for performance evaluation.

Validation Strategy

The model was evaluated using unseen employee-project combinations.

This approach ensures that performance metrics reflect real-world behavior rather than memorization.

Evaluation Metrics

Several industry-standard metrics were used.

Confusion Matrix

Visualizes:

- True Positives
- True Negatives
- False Positives
- False Negatives

These metrics collectively provide a comprehensive understanding of model performance.

6. Deployment and Prediction through the Web Application

After successful training and validation, the machine learning model was integrated into a web-based application.

The web application allows end users to interact with the recommendation engine without requiring any knowledge of machine learning.

User Workflow

Step 1:Project Description Upload,

The user uploads a project description document in PDF format.

This document typically contains:

- Project objectives
- Technical requirements
- Required technologies
- Expected responsibilities

Step 2: Text Extraction

The system extracts textual content from the uploaded PDF.

This converts unstructured project documentation into machine-readable text.

Step 3: Skill Identification

The extracted text is analyzed to identify:

- Required skills
- Preferred technologies
- Project domain knowledge

These skills form the basis of the recommendation process.

Step 4: Employee Data Retrieval

The system loads employee reference information, including:

- Skills
- Experience
- Availability
- Historical performance

Step 5: Feature Construction

For every available employee, the same engineered features used during training are generated. This ensures consistency between training and prediction environments.

Step 6: Model Prediction

The trained machine learning model evaluates every employee-project pair and generates suitability predictions. Each employee receives a recommendation score representing the likelihood of being a strong match for the project.

Step 7: Employee Ranking

Employees are ranked according to their predicted suitability scores. The highest-ranking employees appear at the top of the recommendation list.

Step 8: Results Visualization

The web application presents the results through:

- Top recommended employees
- Match percentages
- Skill comparison charts
- Experience analysis
- Project skill coverage metrics
- Interactive graphical visualizations

This allows project managers to quickly understand why specific employees were recommended.

Conclusion

The HAF-EPA project successfully combines data engineering, machine learning, and web application development to create an intelligent employee recommendation system. By leveraging employee skills, project requirements, historical performance data, availability information, and collaboration metrics, the system provides accurate and explainable employee recommendations.

The solution demonstrates how machine learning can support organizational decision-making, improve resource allocation, reduce manual effort, and increase the likelihood of successful project outcomes. It serves as a practical example of applying predictive analytics and artificial intelligence to real-world workforce management challenges.